Register Number | | | | | | | | | | | | | | | |

PSNA College of Engineering and Technology, Dindigul - 624 622.

(An Autonomous Institution Affiliated to Anna University, Chennai)

End Semester Examinations :: Nov/Dec 2025

B.E	Biomedical Engineering			Max. Marks : 100
Semester:	III	**251106**	**R-2022**	Date : 24.11.2025
BM2312	Electronic Devices and Circuits			Time : 3 Hours

Part - A : Answer All Questions. **10 x 2 = 20 Marks**

		Marks
1	Define barrier potential and depletion region of a PN junction diode.	2
2	Differentiate between diffusion and transition capacitance in a diode.	2
3	Compare the differences between BJT and MOSFET.	2
4	Recall any one application for each BJT and UJT.	2
5	Outline the significance of frequency analysis in amplifier design.	2
6	Compare voltage gain and current gain in CE and CB amplifiers.	2
7	What is the role of a differential amplifier in communication circuits?	2
8	Outline the term "power amplifiers".	2
9	What are the conditions required for oscillation?	2
10	Differentiate between positive and negative feedback amplifiers.	2

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Part - B : Answer All Questions. **5 x 13 = 65 Marks**

11	A	Explain the structure, working principle, and VI characteristics of a PN junction diode.	13
		OR	
	B	Describe the construction, working, and characteristics of a Schottky barrier diode.	13
12	A	Explain the structure and operation of BJT in any one configuration with input and output characteristics.	13
		OR	
	B	Explain the construction and working of UJT and its V-I characteristics.	13
13	A	Derive the small signal model for a CE amplifier and obtain expressions for voltage gain and input impedance.	13
		OR	
	B	Illustrate the various analysis of CS MOSFET amplifier.	13

14 A Explain the operation of a differential amplifier and derive expressions for CMRR. 13

OR

B Outline the operation of different classes of power amplifiers and their efficiency. 13

15 A Outline the types of negative feedback topologies with neat block diagram (or) circuit. 13

OR

B Describe the working principles of Wien bridge and crystal oscillators with circuit diagrams. 13

Part – C : Answer All Questions. **1 x 15 = 15 Marks**

16 A Design a full-wave rectifier circuit using a center-tapped transformer. Analyze its performance parameters with input and output waveforms. Suggest modifications to improve its performance. 15

OR

B Examine the construction and working of LED, LCD and solar cell. 15

